

```
!pip install -U crewai google-generativeai
```

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Requirement already satisfied: crewai in /usr/local/lib/python3.12/dist-packages (0.201.1)
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```

```
from crewai import LLM

llm = LLM(
    model="gemini/gemini-2.5-flash", # LiteLLM-compatible format
    api_key="AIzaSyDKS1N07Z91j2Ig4uG5hLChvafBC6SKWTQ"
)
```

```
from crewai import Agent

summary_agent = Agent(
    role="Product Summarizer",
    goal="Summarize the product features.",
    backstory="Expert in product analysis.",
    llm=llm,
    verbose=True
)

sentiment_agent = Agent(
    role="Review Analyst",
    goal="Analyze customer reviews and extract sentiment insights.",
    backstory="Specialist in understanding customer feedback.",
    llm=llm,
    verbose=True
)

decision_agent = Agent(
    role="Purchase Advisor",
    goal="Make a recommendation on whether to buy the product.",
    backstory="Expert in consumer decision-making.",
    llm=llm,
    verbose=True
)
```

```
from crewai import Task

summarize_task = Task(
    description="Summarize the product: {product_query}",
    expected_output="A detailed summary of the product's features, pros, and cons.",
    agent=summary_agent
)
```

```
sentiment_task = Task(
    description="Analyze the sentiment of these reviews: {review_text}",
    expected_output="A sentiment summary of the reviews.",
    agent=sentiment_agent
)

decision_task = Task(
    description="Make a YES or NO decision on whether to buy the product, based on the product summary and sentiment analysis from
    expected_output="A definitive YES or NO recommendation to buy the product, with a brief justification based on the provided su
    agent=decision_agent,
    context=[summarize_task, sentiment_task]
)
```

```
from crewai import Crew

crew = Crew(
    agents=[summary_agent, sentiment_agent, decision_agent],
    tasks=[summarize_task, sentiment_task, decision_task] # Ensure tasks are in logical order
)

# Input data
product_query = "iPhone 15"
review_text = "Battery life is great, camera is excellent, but it's a bit expensive. Some users say it's worth it, others prefer An

# Run the crew - Agents will use their LLMs based on task descriptions and context
result = crew.kickoff({
    "product_query": product_query,
    "review_text": review_text,
})
```

🤖 Agent Started

Agent: **Product Summarizer**
Task: Summarize the product: iPhone 15

✅ Agent Final Answer

Agent: **Product Summarizer**
Final Answer:
The iPhone 15 introduces several key upgrades and features, building upon Apple's smartphone legacy while making premium features more accessible to the standard lineup. It offers a blend of performance, photography enhancements, and design refinements.

Key Features:

1. **Dynamic Island:** Previously exclusive to Pro models, the Dynamic Island now replaces the notch on the iPhone 15 and iPhone 15 Plus. It fluidly expands and contracts to show important alerts, notifications, and activities, such as Face ID authentication, incoming calls, timer countdowns, and Live Activities for apps like music playback or navigation.
2. **Advanced Camera System:**
 * **48MP Main Camera:** The biggest camera upgrade for the standard iPhone in years, it features a 48-megapixel main sensor with a brighter f/1.6 aperture and sensor-shift optical image stabilization. This allows for incredibly detailed photos, improved low-light performance, and the ability to capture 2x optical quality telephoto shots by cropping from the center of the high-resolution sensor.
 * **Next-generation Portraits:** The camera now automatically detects when there's a person, dog, or cat in the frame and captures depth information, allowing users to instantly turn photos into stunning portraits with beautiful bokeh, or adjust the focus point even after the shot has been taken.
 * **Photonic Engine:** Enhances mid-to-low light performance for photos across all cameras, delivering brighter colors and more detail.
 * **12MP Ultra Wide Camera:** Offers a wider field of view for landscape and group shots.
 * **12MP TrueDepth Front Camera:** Delivers sharp selfies and improved low-light performance.
3. **A16 Bionic Chip:** Borrowed from the iPhone 14 Pro models, the A16 Bionic chip powers the iPhone 15. It features a 6-core CPU, 5-core GPU, and 16-core Neural Engine, providing excellent performance for demanding tasks, gaming, and advanced computational photography.
4. **Brighter Super Retina XDR Display:** The 6.1-inch display (or 6.7-inch for iPhone 15 Plus) is now significantly brighter, reaching up to 1000 nits typical maximum brightness and up to 2000 nits peak outdoor brightness, making content easier to view in direct sunlight. It still maintains the same 60Hz refresh rate.
5. **USB-C Connectivity:** A monumental shift, the iPhone 15 adopts a USB-C port for charging, data transfer, and connecting accessories, replacing the proprietary Lightning port. This brings greater compatibility with a wider range of devices and chargers. However, the port operates at USB 2.0 speeds (up to 480 Mbps), similar to Lightning.
6. **Durable Design:** Features a new color-infused glass back, which is durable and comes in five new frosted matte finishes. The front is protected by Ceramic Shield, which Apple claims is tougher than any smartphone glass. It also maintains IP68 water and dust resistance.
7. **Enhanced Connectivity:** Includes a second-generation Ultra Wideband chip for improved "Find My" precision for friends, Wi-Fi 6, and 5G cellular connectivity.
8. **Safety Features:** Continues to offer vital safety features like Emergency SOS via Satellite and Crash Detection, which can automatically connect users to emergency services when needed.
9. **iOS 17:** Ships with the latest version of Apple's mobile operating system, introducing new features like Contact Posters, StandBy mode, enhanced AirDrop, and Journal app.

Pros:

* **Dynamic Island for All:** Brings a key Pro-level feature to the standard models, enhancing user interaction and aesthetic appeal.
* **Significant Camera Upgrade:** The 48MP main sensor, 2x telephoto option, and improved low-light capabilities offer a substantial leap in photo quality for non-Pro iPhones.
* **Faster Performance:** The A16 Bionic chip ensures smooth operation, excellent gaming, and efficient multitasking.

```
print("\n--- Final Decision ---\n")  
print(result)
```

Improved Design: New color-infused glass back offers a fresh look and feel, while Ceramic Shield ensures durability.

---*Final Decision*---
* **Excellent Battery Life:** Offers all-day battery life, with the iPhone 15 Plus providing even longer endurance.

The iPhone 15 presents a compelling package with significant upgrades that bring premium features to the standard model. Its adoption of USB-C is a major step forward, and the inclusion of Dynamic Island for all models is a welcome move.

* **No ProMotion (60Hz Display):** The standard 60Hz refresh rate can feel less smooth than the 120Hz ProMotion displays on competitor phones and Apple's Pro models, especially when scrolling.
* **USB 2.0 Speeds for USB-C:** While the port is USB-C, the data transfer speeds are limited to USB 2.0, which is disappointing compared to the USB 3 speeds (up to 10Gbps) offered on the iPhone 15 Pro models.
* **No Always-On Display:** Another Pro-exclusive feature that is absent, meaning the screen does not show time, widgets, or notifications at a glance when locked.
* **Incremental Design Changes (Externally):** Apart from the Dynamic Island, the overall design language is very similar to its predecessors, potentially feeling less "new" to some users.
* **No Dedicated Telephoto Lens:** While it offers 2x optical quality zoom, it's achieved digitally by cropping the 48MP main sensor, not via a separate optical zoom lens like on Pro models.
* **Price:** Remains a premium-priced device, which may be a barrier for some consumers, despite offering more value than previous base models.

🤖 Agent Started

Agent: **Review Analyst**

Task: Analyze the sentiment of these reviews: Battery life is great, camera is excellent, but it's a bit expensive. Some users say it's worth it, others prefer Android.

✅ Agent Final Answer

Agent: **Review Analyst**